

Edutech Solution

AI & ML INTERNSHIP

Task 2: Data Cleaning & Missing Value Handling

Tools:

- Python(Pandas, NumPy, Matplotlib/Seaborn)

Dataset:

- "TitanicDataset" or "Housing Prices Dataset"

Hints / Mini Guide:

1. Identify missing values using `.isnull().sum()`.
2. Visualize missing data using heatmaps.
3. Handle missing values: Deletion (Dropna) vs Imputation (Mean/Median/Mode).
4. Use SimpleImputer from Scikit-learn for automated handling.
5. Treat outliers that might be causing data noise.
6. Verify the dataset is clean after processing.

Deliverables:

- Cleaned CSV file
- Python code demonstrating cleaning techniques

Final Outcome:

Internmasterstheart of data preprocessing and quality assurance.

Interview Questions Related To Above Task:

- What are the different types of missing data (MCAR, MAR, MNAR)?
- When is it better to drop rows vs imputing values?
- How do outliers affect missing value imputation?
- What is the difference between Mean and Median imputation?
- What is the role of data cleaning in the ML pipeline?

Submission Guidelines

- **Time Window:**

You can complete the task any time between 10:00AM to 10:00PM on the given day. Submission link closes at 10:00PM.

- **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. [cite: 35, 36] Just learn the process or find free alternatives.

- **GitHub Submission:**

Create a new GitHub repository for each task. Add everything you used for the task - code, datasets, screenshots (if any), and a short README.md explaining what you did.

- **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link provided.

URL : https://docs.google.com/forms/d/e/1FAIpQLSf_3RtphYvxqIgBmtDQLoiBBN6lSKrYX25WdnzXsSA38_ICYg/viewform?usp=dialog

Best of Luck